Examiner only

6. (a) The spectrum from a star is crossed by dark lines. This is called an absorption spectrum.

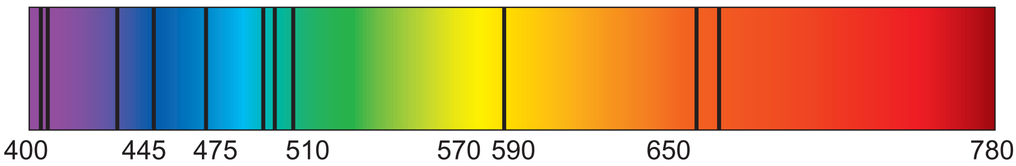
(i) Explain how an absorption spectrum is produced. [2]

....

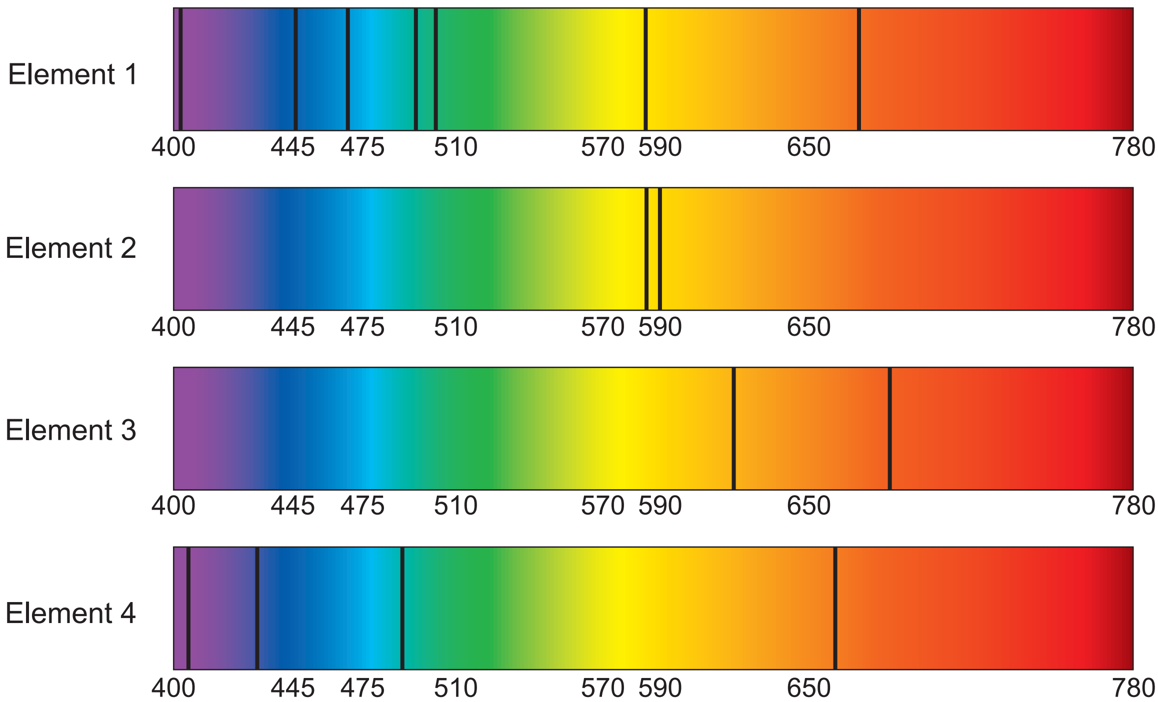
....

....

(ii) An absorption spectrum from a star is shown below.



Here are absorption spectra for different elements.



Examiner only

Complete the table to show whether each element is present in the star. [2]

Element	Present in the star (Yes or No)
1	....
2	....
3	....
4	....

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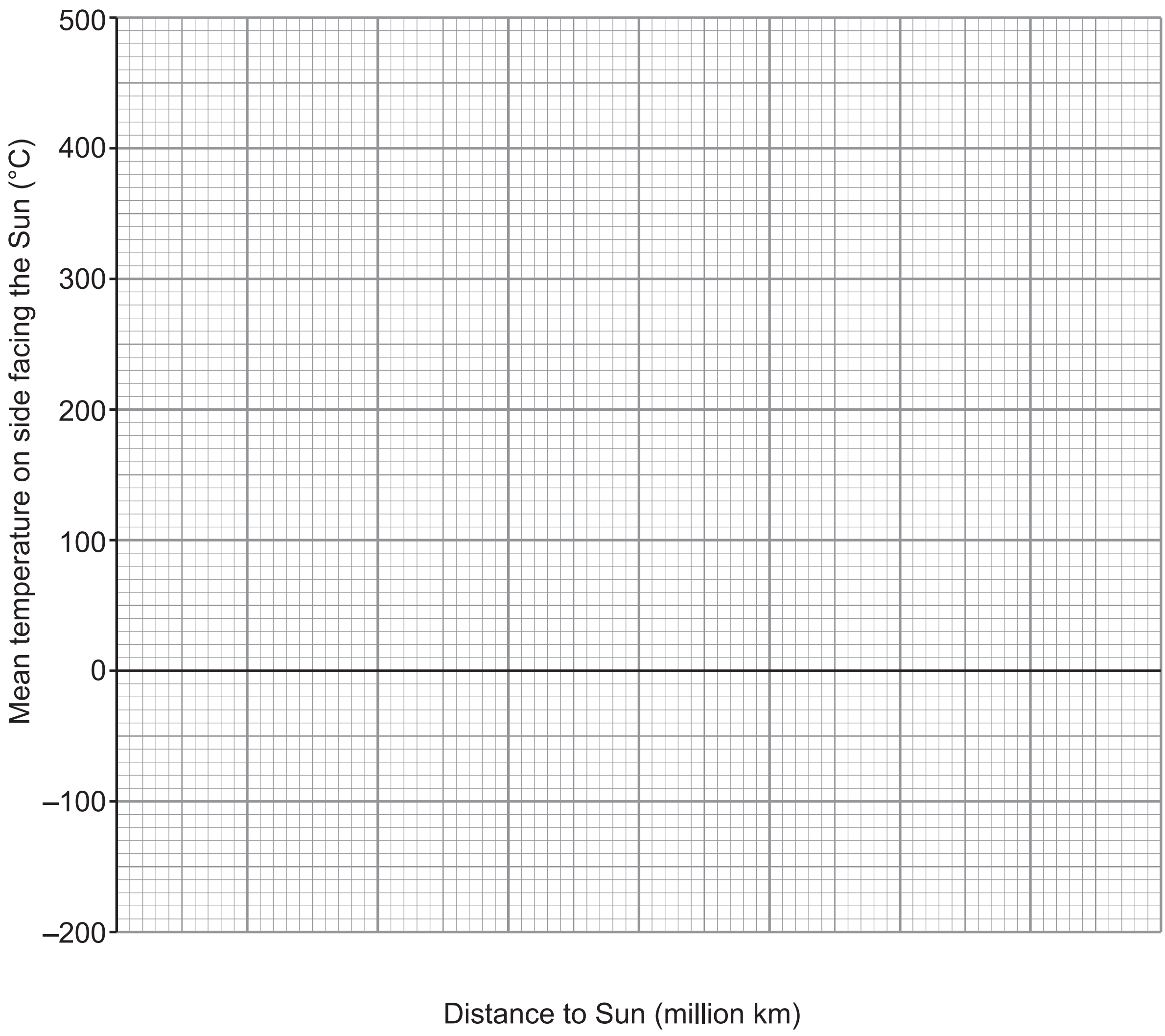
(b) The table gives information about some planets in our solar system.

Planet	Distance to Sun (million km)	Mean temperature on side facing the Sun (°C)
Mercury	60	430
Venus	110	470
Earth	150	20
Mars	230	−20
Jupiter	780	−150

(i) Plot the data on the grid below. [2]



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(ii) Circle the anomalous point on the grid. [1]

(iii) Join the points with a smooth curve. [1]

(iv) Determine the distance from the Sun at which a planet would have a mean temperature to be 0°C. [1]

distance = ... (million km)

(c) Complete the diagram below to show the shape of the orbit of a comet around the Sun. [1]

Sun

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